Anas Mourahib

Postdoctoral Researcher and Lecturer in Statistics

Causal Inference, Graphical Models, Spatial Statistics, and Multivariate Extremes

Personal Information

Date of birth	30 August 1998
Place of birth	Kenitra, Morocco
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Research Profile

Postdoctoral researcher and lecturer in applied mathematics and statistics, specializing in causal inference, graphical models, spatial and spatio-temporal statistics, and multivariate extremes. My research combines structural causal models and the potential-outcomes framework with statistical learning methods for complex real-world data. I currently work on root-cause analysis in collaboration with ASML, a world-leading semiconductor manufacturer, and on the causal effect of protected areas on conflict in Madagascar. I am also a lecturer for the course *Dependence Modeling: Graphical Models and Causality*. My broader interests include scalable methods for assessing interventions with observational spatial and spatio-temporal data.

Education

2021–2025 **PhD in Statistics**, *Université catholique de Louvain (UCLouvain)*, Belgium
Thesis: *Multivariate threshold exceedance approaches in the presence of multiple extreme directions and applications*. Supervised by Prof. Anna Kiriliouk and Prof. Johan Segers.

2020–2021 **Master's degree in Mathematics, Probability, and Statistics**, *Université de Rennes 1*, France

2019–2020 **Maîtrise in Fundamental Mathematics**, *Université Grenoble Alpes*, France

2016–2019 **Bachelor's degree in Mathematics**, *Université Ibn Tofail*, Morocco

Professional Experience

2025–present **Postdoctoral Researcher and Lecturer in Statistics**, *Eindhoven University of Technology*, the Netherlands
Research on causal inference, extreme-value theory and graphical models, including methodological work in statistical machine learning and applications to industrial data. Lecturer for *Dependence Modeling: Graphical Models and Causality*; instructor for *Survival Analysis* and *Applied Biostatistical Modeling*.

2021–2025 **Researcher and Teaching Assistant**, *Institute of Statistics, Biostatistics and Actuarial Sciences (ISBA), UCLouvain*, Belgium
Research on extreme-value theory and graphical models. Instructor for *Probability* and *Statistical Modeling*.

Research Interests

- Causal inference and root-cause analysis
- Causal inference for spatial conflict data
- Treatment-effect estimation and intervention assessment
- Extreme-value theory and risk modeling
- Probabilistic graphical models
- Applications: industry (collaboration with ASML), economics (dependence between the market, sectors and stocks of the S&P 500), hydrology (river discharge data)

Publications

Published and Accepted

2026 **Mourahib, Anas**, Kiriliouk, Anna, and Segers, Johan. A penalized least-squares estimator for extreme-value mixture models. *Computational Statistics & Data Analysis*, Article 108460. [GitHub]

2024 **Mourahib, Anas**, Kiriliouk, Anna, and Segers, Johan. Multivariate generalized Pareto distributions along extreme directions. *Extremes*, 28(2), 239–272. [GitHub]

Work in Progress

2026–present **Mourahib, Anas**, Razakason, Faniry, and Fischer, Julia. Estimating the causal effect of protected areas on conflict in Madagascar.

2025–present **Mourahib, Anas**, Mey, Alexander, Echave-Sustaeta, Ignacio, and Pires da Silva Castro, Rui. Root-cause analysis under mechanism shifts.

2024–present **Mourahib, Anas**, Engelke, Sebastian, and Segers, Johan. Extremal graphical models with non-standard extreme directions. Preprint available in Section 4 of my PhD thesis.

Selected Projects

2026–present **Causal Effect of Protected Areas on Conflict in Madagascar**: collaboration with Faniry Razakason and Julia Fischer, started at the StatML–Bocconi Summer School on causality. Madagascar is well known for its biodiversity, and many initiatives protect areas of high biodiversity, for example by creating national parks. However, many local inhabitants depend on these areas for their livelihood, for instance through agriculture or hunting, which can create conflict. The aim of the project is to quantify the causal effect of protected areas on conflict.

2025–present **Root-Cause Detection Project** (postdoctoral research in collaboration with ASML): developing data-driven methods to identify and diagnose root causes in complex industrial systems using statistical learning, causal inference and dependence modeling.

2025–2026 **Dependence Modeling Course Materials**: public repository containing simulations and applied examples in R and Python on structure learning for undirected and directed graphical models, with an emphasis on causal modeling. Code: GitHub.

2024–2025 **Penalized Least-Squares Estimator for Extreme-Value Mixture Models**: public repository accompanying the related publication, with implementations in R, C, Python and Julia. Code: GitHub.

Honors and Awards

2023 FNRS grant for international mobility

Academic Service and Organization

7–9 April 2027 Co-organizer of the EURANDOM Workshop on Probabilistic Graphical Models, with Carlos Améndola, Ignacio Echave-Sustaeta Rodríguez, Francesco Nowell, and Frank Röttger.

Research Stays

February–May 2024 Research Center for Statistics, University of Geneva. Collaboration with Prof. Sebastian Engelke.

Computer Skills

R Advanced proficiency; reproducible workflows with R Markdown; version control with Git and GitHub

Python Advanced proficiency; Jupyter notebooks; scikit-learn, PyTorch, and TensorFlow; version control with Git and GitHub

Julia Advanced proficiency

C/C++ Advanced proficiency

LaTeX Working proficiency

Languages

Arabic	Native or bilingual proficiency
Dutch	Elementary to intermediate proficiency (A2/B1)
English	Full professional proficiency
French	Full professional proficiency

References

Rui Pires da Silva Castro	Associate Professor of Statistics, Department of Mathematics and Computer Science, Eindhoven University of Technology. r.m.pires.da.silva.castro@tue.nl
Frank Röttger	Assistant Professor of Statistics, University of Twente. f.rottger@utwente.nl
Anna Kiriliouk	Assistant Professor of Statistics, Institute of Statistics, Biostatistics and Actuarial Sciences (ISBA), UCLouvain, Belgium. anna.kiriliouk@uclouvain.be
Johan Segers	Full Professor of Statistics, Department of Mathematics, KU Leuven, Belgium. jjjsegers@kuleuven.be